



School of Computing

Master of Computer Applications (MCA)

Course Name : Data Warehousing and Mining

Course Code : TMC315

Semester : 3

Pre-requisite : Basic knowledge of database management systems and fundamental statistical concepts.

L	T	P	Credits
3	1	0	4

Course Learning Objectives:

Upon successful completion of this course, students will be able to:

- Understand the fundamental concepts, architecture, and applications of Data Mining and Data Warehousing.
- Apply data preprocessing techniques including cleaning, integration, transformation, and summarization to improve data quality.
- Analyze and implement various data mining techniques such as association rule mining, classification, prediction, and clustering.
- Evaluate different data mining algorithms and their suitability for solving real-world business and research problems.
- Explore advanced data mining concepts including spatial, multimedia, text, web, and time-series data mining.

After completion of the course, the students will be able to:

Course Outcome	Statement	Bloom's Level	Skill
CO1	Explain the concepts of data mining, data warehousing, OLAP, multidimensional data models, and warehouse architectures.	L2 – Understand	Conceptual Understanding, Analytical Thinking
CO2	Perform data preprocessing using techniques such as data cleaning, integration, transformation, reduction, and descriptive summarization to prepare datasets for mining.	L3 – Apply	Data Preparation, Problem Solving
CO3	Apply association rule mining techniques, including the Apriori algorithm and correlation analysis, to discover meaningful patterns from transactional and relational databases.	L4 – Analyze	Data Analysis, Pattern Discovery
CO4	Apply clustering and classification methods to analyze datasets and develop models..	L3 – Apply	Machine Learning, Data Analysis,
CO5	Analyze advanced data mining applications involving spatial, multimedia, temporal, sequence, text, and web data to solve data analysis problems	L4 – Analyze	Critical Thinking, Knowledge Discovery



School of Computing

Master of Computer Applications (MCA)

Details of the Course:

Unit 1: Introduction to Data Mining and Data Warehouse

Contact Hours/Teaching Hours: 12

Fundamentals of Data Mining, Data Mining Functionalities, Classification of Data Mining Systems, Major Issues in Data Mining, Knowledge Discovery in Databases (KDD)

Data Warehousing: Overview, Definition, Components of Data Warehouse, , Difference between Database Systems and Data Warehouse, Data Warehouse Models: Enterprise Warehouse, Data Mart, and Virtual Warehouse, Data Cubes, Concept Hierarchy, Multidimensional Data Model : Star Schema, Snowflake Schema, Fact Constellation, , Three-Tier Architecture, Data Warehouse Measures, Categorization and Computation of Measures, OLAP Operations : Roll-up, Drill-down, Slice, Dice, Pivot, Advantages of OLAP over OLTP, OLAP Architecture : MOLAP, ROLAP , HOLAP

Self-Learning Hours = 6

Students shall explore real-world applications of data mining and data warehousing, study different warehouse architectures and schemas, and investigate OLAP tools and multidimensional data models.

Team Activity Hours = 3

Group discussion and presentation on the applications of data mining and data warehousing in domains such as banking, healthcare, retail, and e-commerce, including comparison of OLTP and OLAP systems.

Unit 2: Data Preprocessing

Contact Hours/Teaching Hours: 12

Need for Data Preprocessing, Descriptive Data Summarization, Data Cleaning: Handling Missing Values, Noisy Data (Binning, Clustering, Regression, Computer and Human Inspection), Inconsistent Data, Data Integration, Data Transformation, Data Normalization, Data Scaling : Min-Max Normalization, Z-score Normalization, Decimal Scaling

Self-Learning Hours = 6

Students shall perform preprocessing tasks on sample datasets using data cleaning, transformation, integration, normalization, and summarization techniques.

Team Activity Hours = 3

Group-based activity on preprocessing real-world datasets and comparing different techniques for improving data quality.

Unit 3 Association Rule Mining and Pattern Discovery

Contact Hours/Teaching Hours: 12

Data Mining Techniques : Market Basket Analysis, The Apriori Principle, Interestingness Measures : Support and Confidence, Frequent Itemset, Mining Association Rule Generation; Apriori algorithm FP-Growth Algorithms, correlation analysis.

Self-Learning Hours = 6



School of Computing

Master of Computer Applications (MCA)

Students shall implement the Apriori algorithm, generate frequent itemsets, and analyze association rules using sample transactional datasets.

Team Activity Hours = 3

Group activity on market basket analysis using association rule mining techniques and interpretation of discovered patterns.

Unit 4: Data Mining Techniques

Contact Hours/Teaching Hours: 12

Classification and Prediction, Issues related to classification & prediction, classification Process, Decision Tree Induction, Naïve Bayes Classification. K-nearest neighbour classifiers.

Introduction to Clustering techniques, Data types in cluster analysis, categories of clustering techniques: Partitioning Methods (K-Means, K-Medoids), and Hierarchical method.

Self-Learning Hours = 6

Team Activity Hours = 3

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Unit 5: Overview of Advanced Features of Data Mining

Contact Hours/Teaching Hours: 12

Mining complex data objects, Spatial databases, Multimedia databases, Time series and Sequence data; mining Text Databases and Mining Word Wide Web

Self-Learning Hours = 6

Students shall study advanced data mining applications and explore case studies involving text, web, spatial, multimedia, and time-series data.

Team Activity Hours = 3

Group presentation on advanced data mining applications in social media analytics, recommender systems, healthcare analytics, fraud detection, and web intelligence.

Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2	PSO3
CO01	3	2	-	2	-	-	-	1	2	-	2
CO02	3	3	2	3	-	-	-	2	3	-	3
CO03	2	3	3	3	-	-	-	2	3	-	3
CO04	2	3	3	3	-	-	1	2	3	1	3
CO05	2	2	2	3	-	-	1	3	2	1	3
Avg.	2.4	2.6	2.5	2.6	-	-	2.0	2.0	2.6	1	2.8

Text Books:

1. Han, J., Kamber, M., & Pei, J. (2012). *Data Mining: Concepts and Techniques* (3rd Edition). Morgan Kaufmann.



School of Computing

Master of Computer Applications (MCA)

2. Paulraj Ponniah (2010). *Data Warehousing Fundamentals: A Comprehensive Guide for IT Professionals* (2nd Edition). Wiley India.

Reference Books:

1. Margaret H. Dunham (2006). *Data Mining: Introductory and Advanced Topics*. Pearson Education.
2. Pang-Ning Tan, Michael Steinbach, & Vipin Kumar (2019). *Introduction to Data Mining* (2nd Edition). Pearson Education.
3. Alex Berson & Stephen J. Smith (2007). *Data Warehousing, Data Mining, and OLAP*. McGraw-Hill Education.

Online Resources:

Online Resources:

1. **GeeksforGeeks – Data Mining Tutorials**
<https://www.geeksforgeeks.org/data-mining/>
2. **NPTEL Courses**
<https://nptel.ac.in/>
3. **IBM – Data Mining Overview**
<https://www.ibm.com/think/topics/data-mining>
4. **Oracle – Data Warehousing Documentation**
<https://www.oracle.com/database/technologies/data-warehouse.html>
5. **Microsoft Learn – Data Analytics and Data Warehousing**
<https://learn.microsoft.com/>
6. **Weka Machine Learning Software** (*Useful for Data Mining Practicals*)
<https://www.cs.waikato.ac.nz/ml/weka/>
7. **RapidMiner Studio** (*Data Mining Tool*)
<https://rapidminer.com/>
8. **Kaggle – Datasets for Data Mining Practice**
<https://www.kaggle.com/datasets>
9. **UCI Machine Learning Repository**
<https://archive.ics.uci.edu/ml/index.php>
10. **Scikit-learn Documentation** (*Machine Learning & Data Mining Algorithms*)
<https://scikit-learn.org/stable/>